

PD Group:

Name:

ANDERSON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS 2008

BIOLOGY

Higher 1

Paper 2 Core Paper

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READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graph or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

All working for numerical answers must be shown.

Section B

Answer any **one** question.

Write your answers on separate writing paper.

At the end of the examination,

1. fasten all your work securely together;
2. hand in your answers for Section A and Section B separately.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

Section A

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3

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Section B

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Section A

Answer all the questions in this section.

QUESTION 1

Fig. 1.1 is an electron micrograph of a fibroblast. Fibroblasts are flattened, irregular-shaped connective tissue cells that secrete components of extracellular matrix such as hyaluronic acid and collagen.

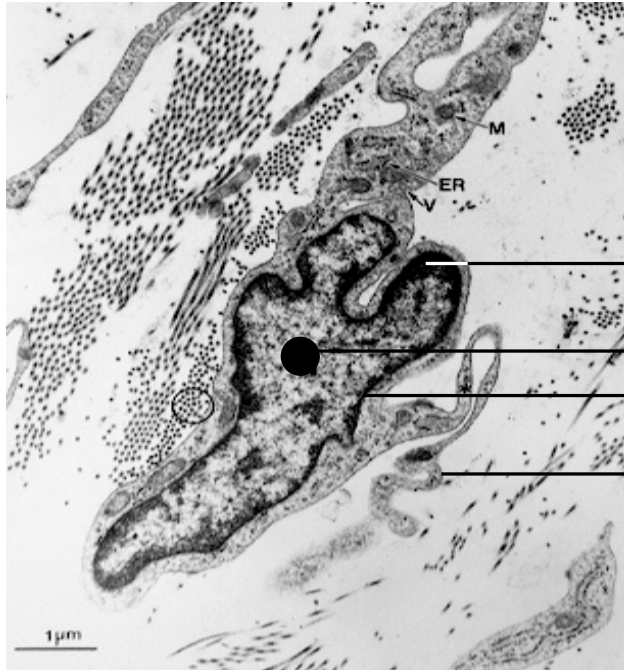


Fig. 1.1

Picture adapted from www.udel.edu/~histopage

(a) Name structures A to D.

A

B

C

D

[2]

(b) Explain why collagen is described as a fibrous protein.

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B

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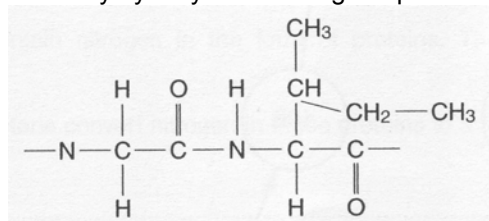
[2]

Collagen is continuously broken down in the extracellular matrix by the enzyme collagenase, which catalyses the hydrolysis of the peptide bond between the amino acids glycine and isoleucine.

(c) Suggest how collagenase is only able to act on the peptide bond between glycine and isoleucine and not on peptide bonds between any other amino acids.

[2]

(d) Draw a diagram below to show how the peptide bond between glycine and isoleucine is broken down by hydrolysis including the product or products.



(e) Suggest why collagen has great tensile strength.

[1]

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QUESTION 2

When a red flowered, three-leaved clover plant was crossed with a white flowered, five-leaved plant the following offspring were produced:

red-flowered, five leaved	36
red-flowered, three-leaved	41
white-flowered, five-leaved	40
white-flowered, three-leaved	38

The alleles for red flowers and five leaves are dominant.

- (a) Using the symbols **R** for red flowers, **r** for white flowers, **L** for five leaves and **l** for three leaves, draw a genetic diagram to explain these results. [5]

Deleted: Note : Half life refers to the time taken for the amount of mRNA to be degraded to half the original amount

¶

¶

(d) (i) Explain why luciferase activity can be used as a reflection of translation efficiency. [2]

¶

<#>With reference to the Table 3.1, describe the effect of polyadenylation on mRNA degradation in capped mRNA. [2]

¶

<#>

<#> (iii) With reference to Table 3.1, describe the effects of capping and polyadenylation on translation efficiency. [2]

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(d) (i) Luciferase (protein) is the end product of translation; easily detected as a light / measurable [2]

¶

(ii) Increase/ doubles mRNA half-life; from 53 units to 100 [54]

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A cross was made between two [58]

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(b) When these plants were self-pollinated, explain why only the white-flowered, three-leaved plants bred true.

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<#>Whether observed results differed significantly from expected results with a less than 5%;¶

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Vestigial wing and ebony bodied flies would be disadvantaged in the wild because they cannot fly from and are more visible to predators and therefore unlikely to survive as they are more likely to be eaten. [66]

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The inheritance of coat colour in a type of leopard was determined to be due to two gene loci each with alleles which have a dominant-recessive relationship. Leopards with black coats, white coats or spotted coats have been observed.

When a pure-breeding white leopard was crossed with pure-breeding black leopards, the F₁ progeny obtained were all spotted. When the spotted progeny were mated with each other, the results of the F₂ generation were

93 spotted
44 black
28 white

The suggested pathway for the synthesis of pigments involved in the development of coat colour is shown below.

Black pigment $\xrightarrow{\text{Enzyme A}}$ White pigment $\xrightarrow{\text{Enzyme B}}$ Spot formation

(c) Explain the results of the F₂ generation.

[4]

[Total: 11]

QUESTION 3

One way of investigating relationships between different primate groups is to study differences in the amino acid sequences of proteins such as haemoglobin.

(a) Explain the origin of differences in the amino acid sequences in a protein.

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The β chain of haemoglobin has been investigated and shown to have only four points of difference in all primate groups. The amino acids found at these positions in certain primates or primate groups are shown in the table below.

Primate or primate group	Amino acid at position 80	Amino acid at position 87	Amino acid at position 109	Amino acid at position 125
Chimpanzees	Aspartic acid	Threonine	Arginine	Proline
Gibbons	Asparagines	Lysine	Arginine	Glutamine
Old World monkeys	Aspartic acid	Glutamine	Lysine	Glutamine
New World monkeys	Aspartic acid	Glutamine	Arginine	Glutamine
Lemurs	Aspartic acid	Glutamine	Threonine	Alanine

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Table 3.1

(b) (i) With reference to Table 3.1, state the two groups that seem to be closely related and give reasons for your answer.

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[2]

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(ii) Gibbons are usually classified with the chimpanzees as apes, in the Hominoidea. Explain why the results shown do not support this classification.

[3]

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6 In a recent study, a family (Feline) was found to be extinct. Within a few pockets of the population, it was suggested that this species in population most populated species at a very high.

i. Species

ii. Genet

(b) What impact species survival

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(a) caused by point mutation (... [77]

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Only one difference / three co (... [78]

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Gibbon β chain has only one (... [79]

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(... [80]

(b) Draw (in Fig. 4.2) the bands that would be obtained for

(i) Lane N: fetus which is homozygous for the diseased allele

(ii) Lane O: fetus which is heterozygous for the diseased allele

(c) State how doctors analyze the DNA fragments to identify the fetus with the genetic defect.

[Total: 8]

(ii) Lane O:
(c) State how (defect.[1]

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insulin molecule

cell surface membrane

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Section B: Free Response Questions
Answer only **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

5. (a) Outline the role of NAD and NADP in cells.
(b) Describe how photophosphorylation differs from oxidative phosphorylation.

[Total: 20]

6. (a) Describe the structure and function of the portions of eukaryotic DNA that do not encode for protein or RNA.
(b) Explain how the substitution of one nucleotide in DNA can be selected for or against in an area endemic with malaria.

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Nov 2003 Paper 2 Q1

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**Insoluble in water;
Secondary structure (helical);
Polypeptide chains cross link to form long fibres;
Regular amino acid sequence / every 3rd residue is a glycine**

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Collagen is found in the extracellular matrix of muscles, tendons, ligaments and bones. Fibroblast cells in these tissues make collagen by synthesizing polypeptides that form molecules with a triple helix shape. These are secreted from fibroblasts into the extracellular matrix where enzymes assemble them into collagen fibres.

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VJC Promo 2006 Q1d

(a) State two similarities between rough endoplasmid reticulum and golgi apparatus.

[
2
]

(b) State two differences between rough endoplasmid reticulum and golgi apparatus.

[2]

Similarity (<i>any one</i>) - Both consist of membrane-bound sacs/tubules called cisternae. [1] - Both are made up of single membrane; [1] - Both produce vesicles that bud off the organelle; [1/2]	
Differences	
Structure D (rER)	Structure F (GA)
1. Consist of a <u>network</u> of <u>interconnected</u> membrane-bound sacs and tubules. [1/2]	Consist of <u>stacks</u> of <u>flattened</u> membrane-bound sacs. [1/2]
NB: idea of interconnectedness is imp.	
2. Presence of ribosomes. [1/2]	Absence of ribosomes. [1/2]
® ribosomes <i>may or may not</i> be present	
3. Connected to the outer membrane of the nucleus. [1/2]	3. Not connected to the outer membrane of the nucleus. [1/2]
4. Does not have a definite forming and maturing face.	4. Has a definite forming face where the vesicles join and a maturing face where the vesicles bud off.

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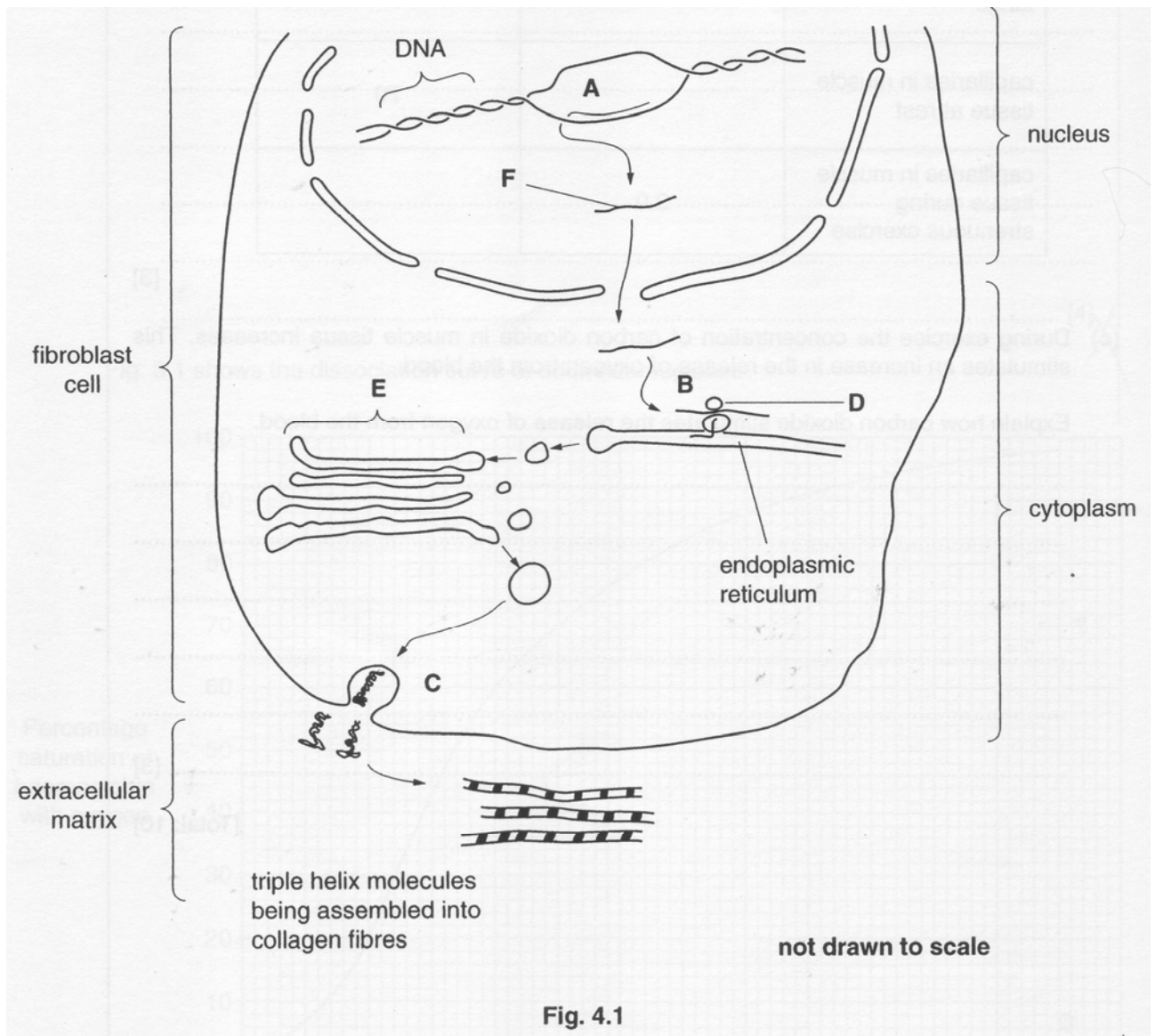
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Fig. 4.1 is a diagram summarizing these events.



(a) (i) Name the processes occurring at **A**, **B** and **C**. [3]

A Transcription
B Translation
C exocytosis

(ii) Name structures **D** and **E**. [2]

D Sub unit of ribosome
E Golgi apparatus/body

(iii) Name molecule **F**. [1]

F mRNA

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a- to remove diagram and combine with something else....		

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Nov 2003 Paper 2 Q1

(c) Explain why collagen is described as a fibrous protein.
[2]

insoluble nature of collagen;
secondary structure – three polypeptides chains are cross-linked via
hydrogen bonding to each other to form a triple helix;
Fibrils like parallel in bundles to form a collagen fiber;

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b

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active site;
specific complementary shape
only accepts R grps of these 2 amoni acids

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Correct bond broken (between C-N)
Involvement of water molecules in breaking the peptide bond shown clearly
Two amino acids with free grps as follows
-COOH/-COO⁻ and -NH₂/-NH₃⁺

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b and c are good :)

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VJC Promo 2006 Cell and Nuclear Division

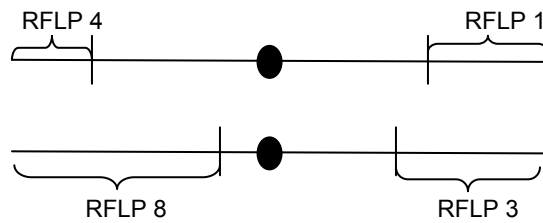
Cell and Nuclear Division

RJC Prelim 2007

QUESTION 2

Huntington's Disease (HD) is a neurodegenerative disorder due to an autosomal dominant allele. The disease is characterized by progressive motor dysfunction and intellectual deterioration but its symptoms usually occur when individuals are in their 30's-50. To circumvent the problem, RFLP has been used these days for early detection of the disease.

Patients with HD are known to have RFLP 1 and the diagram below shows a pair of homologous chromosomes from a male.



An offspring of the male has a chromosome with RFLPs 1 and 8, contributed by the father.

- (a)(i) State the event that must occur during meiosis in order to obtain a chromosome with RFLPs 1 and 8.
[1]

-
- (ii) In the space below, draw the event occurring in (a).
[3]

- (iii) Suggest a reason why Huntington's Disease is likely to persist in populations. [1]

The micrograph below shows a Lily cell undergoing cell division.



[© Clare Hasenkampf/BPS]

- (b) Refer to the micrograph above and:
- (i) Identify the type and stage of nuclear division.
- [2]

(ii) Briefly explain the reason for your answer in (b)(i).
[2]

(c) Describe the roles of spindle fibers in meiosis.
[2]

Tubulin is involved in the construction of microtubules.

(a) State where microtubules originate in the cell.

_____ [1]
]

(b) Describe the role of microtubules in mitosis.

_____ [1]
]

The chromosome drawings show four stages in mitosis.

W

X

Y

Z



(c) Give the correct sequence of the drawings.

_____ [1]
]

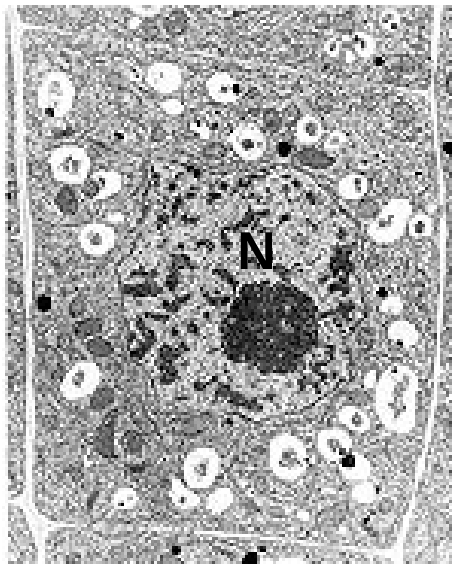
(d) What is the diploid number of the chromosome in these cells.

_____ [1]
]

(e) If you look at Figure Z, the chromosomes appear to be double structures. Suggest why?

_____ [1]

The DNA in the nucleus (**N**) of the plant cell in the electron micrograph, Figure 5.1, below appears different from that of a cell undergoing mitosis (Figure 5.2).



(http://www.museum.vic.gov.au/scidiscovery/images/mn003613_w150.jpg)
 (http://plantphys.info/Plant_Biology/images/anaphase1.jpg)

Figure 5.1



Figure 5.2

Account for this difference.

[3

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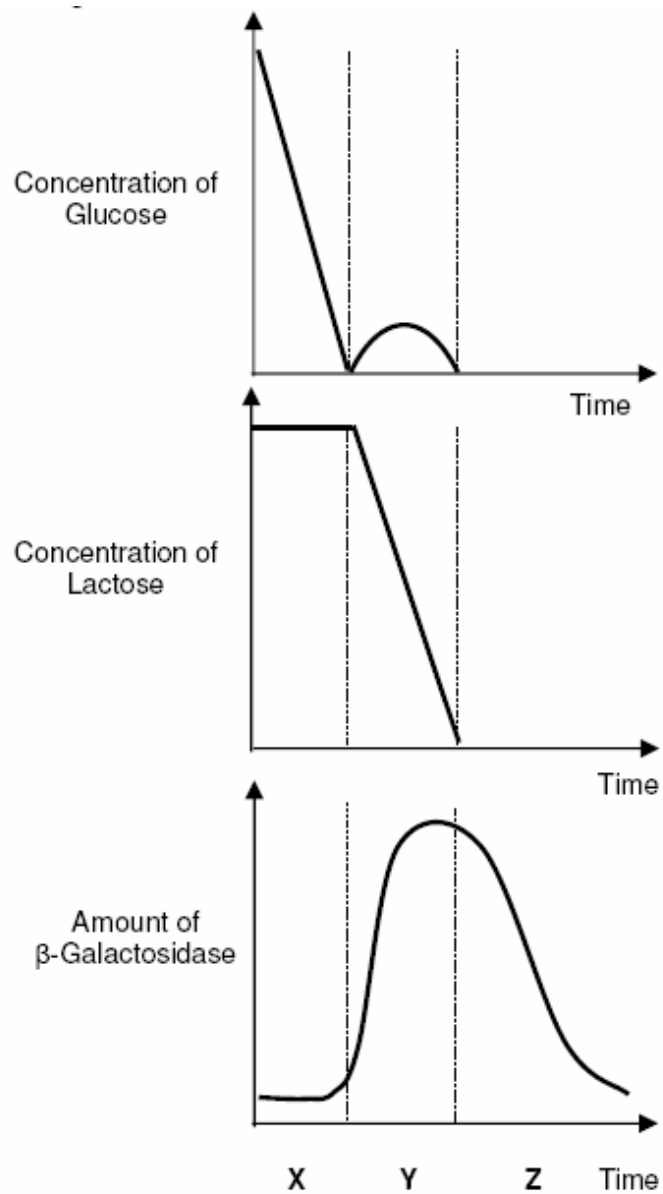


Fig.1

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what is meant by "dual control of the <i>lac</i> operon"		
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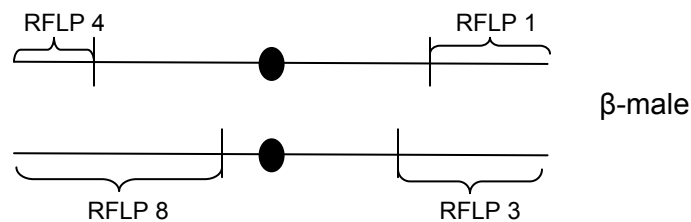
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AJC

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Huntington's Disease (HD) is a neurodegenerative disorder due to an autosomal dominant allele. The disease is characterized by progressive motor dysfunction and intellectual deterioration but its symptoms usually occur when individuals are in their 30's-50. To circumvent the problem, RFLP has been used these days for early detection of the disease.

Patients with HD are known to have RFLP 1 and the diagram below shows a pair of homologous chromosomes from a β -male.

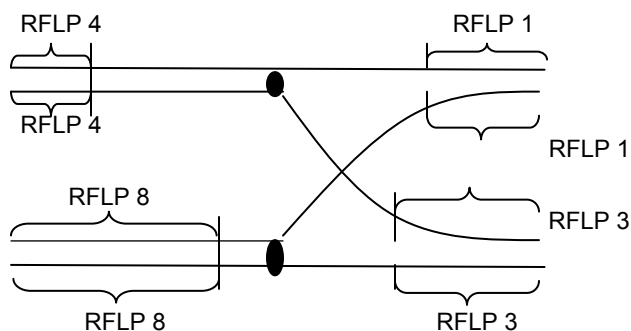


An offspring of β -male has a chromosome with RFLPs 1 and 8, contributed by the father.

(a)(i) State the event that must occur during meiosis in order to obtain a chromosome with RFLPs 1 and 8. [1]

crossing over

(ii) In the space below, draw the event occurring in (a). [3]



[1] - 2 sister chromatids per chromosome
[1] - a crossing over of non-sister chromatids; may also take place on the other arm between RFLP 4 and RFLP 8
labelling of centromere (1/2) and chromatids/chromosomes (1/2)

- (iii) Suggest a reason why Huntington's Disease is likely to persist in populations [1]

Because of its late onset

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The micrograph below shows a lily cell undergoing cell division.



[© Clare Hasenkampf/BPS.]

- (b) In reference to the micrograph above, identify:

- (i) The type and stage of nuclear division [2]

Meiosis

- (ii) Briefly explain the reason for your answer in (b) (i). [2]

Metaphase bec:

metaphase bec chromosomes are aligned at equator

metaphase 2 bec:

2 metaphase plate has formed → meiosis

OR

the daughter cell have chromosomes aligned at the new equatorial plate which is at right angles to the first

- (c) Describe the roles of spindle fibres in meiosis. [2]

pull sister chromatids/homologous chromosomes to opposite poles at anaphase

help align chromosomes at equatorial plate

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Other (non-kinetochore) spindle fibres lengthen to elongate the cell in preparation for cytokinesis

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Describe another mechanism which regulates the expression of β -galactosidase at Y.		
[2]		

(d)

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(a) Centrosome / microtubule organizing centre (MTOC) / centrioles; [1]

(b) Form spindle fibres / spindle;

For separating sister chromatids; (**DO NOT** mention "splitting of sister chromatids")

Equal distribution of chromosomes;

(1/2 each; 1 max)

(c) Y, X, Z, W [1]

(d) 6 [1]

(e) During S phase of interphase, replication of DNA occurs;

Condensation of DNA molecules tightly around histones;

Two DNA molecules appear as sister chromatids in prophase;

(1/2 each; 1 max)

(f) Account for this difference. [3] (1/2 each; 3 max)

Figure 5.1:

DNA as chromatin;

Long, thin / not visible;

DNA wound around histones;

Figure 5.2:

DNA replicated during interphase;

Highly condensed DNA by spiralisation;

Chromosome consists of two sister chromatids;

Sister chromatids separate and move, centromeres first, to opposite poles of spindle fibres;

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suggest one mechanism that causes the amount of β -galactosidase to decrease.		

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RJC promo 2006 DNA & Genomics

Genetics of bacteria

TJC Prelim 2007

QUESTION 3

Bacteria can regulate the expression of its genes in response to changes in the environment, typically to the presence or absence of nutrients. Some bacterial genes

encode for regulatory proteins which controls the expression of other genes or operons.

For example, *Escherichia coli* can utilize any sugar as its source of carbon and energy.

However, when grown in a medium containing both glucose and lactose, the bacteria will

use up one before the other. This preferential use of carbon source is said to be governed

by the “**dual control of the *lac* operon**”.

mRNA degradation/ protein is degraded.

[1]

An experiment was conducted to examine the effects of glucose and lactose on the levels

of β -galactosidase in *E.coli*. Lactose and glucose were added to a culture of bacteria at the start of the experiment and the levels of each were measured at specific time intervals.

The results are shown in Fig. 1 below.

Highlight

State the two mechanisms of regulation of *lac* operon

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**negative control by the *lac* repressor;
positive control by cAMP receptor protein;**

[2]

[1 mark each]

(b) With reference to Fig.1, explain how expression levels of β

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-galactosidase is
regulated by the lac repressor from time X to Y.

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X:

Both glucose and lactose are present and only glucose is used;
In the presence of lactose, allolactose binds to lac repressor;
Thus, the Lac repressor is in its inactive configuration and does not
bind to the operator;

But another mechanism prevents transcription of the lac operon due
to catabolite repression (ie. CAP is inactive and disengages from
CAP binding site);

RNA polymerase is unable to bind to the promoter/Low β –
galactosidase mRNA is produced;

[Any 3 - 1 mark each]

Y:

Once the glucose concentration has depleted, *E. coli* uses Lactose;
Allolactose binds to the lac repressor;
Lac repressor is inactive and does not bind to the operator;
RNA polymerase is able to bind to the promoter to transcribe the
mRNA of the lacoperon / Lac operon is switched on / High β –
galactosidase mRNA is produced;

[Any 3 - 1 mark each]

[6]

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X:
Y:

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(c)

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(i) S

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]

(d) (i) Luciferase (protein) is the end product of translation;
easily detected as a light / measurable [2]

(ii) Increase/ doubles mRNA half-life;
from 53 units to 100 units; [2]

(iii) Capping increases translation efficiency between 21 to 297 folds;
Polyadenylation increases translation efficiency between 1.5 to 21
folds [2]

(e) 5 Small nuclear Ribonuclear Proteins (snRNPs) associate with other proteins;
form a **spliceosome**;
RNA-protein interactions create a **lariat** structure;
Spliceosome recognizes the splicing signals/ splice site/ short nucleotide
sequences at the end of intron on a pre-mRNA molecule;
Lariat structure excises introns;
Joins exons together to form mature mRNA;

c and d ii too specific but can use 1st part.

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Genetic Variation (Crossing)
Nov 2006 Paper 2 Q4

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A cross was made between two fruit flies. One parent had an unknown genotype since it had the wild phenotype (normal [long] wings and normal body colour). The other parent had vestigial [undeveloped] wings and ebony body colour, so it was known to be homozygous recessive for both characteristics.

The resulting offspring were as follows:

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normal wings and normal body colour

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normal wings and ebony body colour

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vestigial wings and normal body colour

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vestigial wings and ebony body colour

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Using the following symbols:

N normal wings **n** vestigial wings

E normal body **e** ebony body

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Draw a genetic diagram in the space below to show the cross described.

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Parental phenotype	normal [long] wings and normal body colour	vestigial [undeveloped] wings and ebony body colour
Genotype	Unknown (NnEe)	nnee;
meiosis		

Gametes	NER I NerI nE	rLne ; ;
meiosis		
Gametes	NER I	Ner I nE ne;
ne	NnEe	Nnee nnEe Nnee;
F1 genotype	1 NnEe	1 Nnee 1 nnEe 1 nnee;
F1 phenotype	normal wings and normal body colour	normal wings and ebony body colour vestigial wings and normal body colour vestigial wings and ebony body colour;

Vestigial wing and ebony body are caused by artificially induced mutations in laboratory bred fruit flies. Suggest why such mutants are unlikely to be found in natural populations.

[2]

Vestigial wing and ebony bodied flies would be disadvantaged in the wild because they cannot fly from and are more visible to predators and therefore unlikely to survive as they are more likely to be eaten; Mutations are rare and such alleles are not passed on to offspring because they are unlikely to be reproduced; Such characters are coded for by recessive alleles which are therefore masked by dominant alleles;

Use the chi-squared (χ^2) test, and the table of probabilities shown in Table 1, to find the probability of the results of the cross in (a) departing significantly by chance from expectation. Show your working.

[5]

$$\chi^2 \text{ test} \quad \chi^2 = \sum \frac{(O - E)^2}{E} \quad v = c - 1$$

Key to symbols

Σ = 'sum of ...'

O = observed 'value'

v = degrees of freedom

c = number of classes

Distribution of χ^2

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.60	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.34	16.27
4	7.78	9.49	11.67	13.28	18.46
5	9.24	11.07	13.39	15.09	20.52

Table 1

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Penotype	Observed (O)	Expected (E)	(O-E) ²	(O-E) ² /E
Red flowered, five -leaved	36	38.75	7.5625	0.195
Red flowered,	41	38.75	5.0625	0.131

three - leaved				
White flowered, five -leaved	40	38.75	1.5625	0.040
White flowered, three - leaved	38	38.75	0.5625	0.015
				$\chi^2 = 0.381;$

degrees of freedom = $n - 1$
 $= 4 - 1 = 3;$

Calculated $\chi^2 = 0.381$
Critical $\chi^2 = 7.82;$

Probability that observed results deviated from expected due to chance is high (0.05 level of significance);

Therefore accept null hypothesis. Observed results are not significantly different from expected results;

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Phenotype ration should be 9: 3: 3: 1, but ratio obtained was 9 spotted: 3 white: (3+1) black [1]

A_B _ genotypes give spotted as dominant alleles give functional enzymes A& B = spotted [1]

aa _ _ genotypes remain black because A enzyme is not produced [1]

A_ bb genotypes are white because B enzyme not produced [1]

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two

[4]

Answers

- (a) caused by point mutation;
changes the sequence of DNA bases / changes nucleotide / codon;
new triplet codes for a different amino acid;
substitution causes frameshift;
codes for new polypeptide [2]

- (b)(i) Old World monkeys and New World monkeys;
Only one difference / three common amino acids [2]

- (b)(ii) Gibbon more like New World monkey;
Gibbon β chain has only one common arginine / amino acid with chimpanzee;
2 common amino acids with New World monkeys;
chimpanzee has only common amino acid with / 3 different from each of the other groups [3]

- (c) anatomical comparisons of fossil primates;
 dating of fossils;
 sequencing / profiling DNA / DNA hybridization / ref. to PCR / DNA fingerprinting
 anatomical comparisons of the skeletons of modern species;
 ref to homology / skeletal / skull / dentition details;
 antibody-antigen / immunological / serum studies;
 use of antibodies / agglutination/ immuno-precipitation; [any 2]

modify the peak for fig. 6.2 at 1st part...

6 What do you understand by the following terms?

- (a) i. Species [2]
 - Any group of organisms;
 - Capable or potentially capable of interbreeding with one another;
 - To produce fertile offspring;
 - Show any similarities in evolution, ecology, morphology, behavior, biochemistry (any 2);
- ii. Genetic variation [1]
 - Members of a population;
 - Differ in one or more inherited characteristics;
- (b) What impact would low genetic variability in a population of this species have on species survival? Explain your answer. [2]
 - Low number of alleles would reduce probability of the presence of a favorable characteristic in the population;
 - If there was an environmental change;
 - Reduced chances of survival / species extinction;
 - Continuous inbreeding between closely related individuals;
 - Inbreeding depression may occur;
- (c) Using the information from **Figure 6.1**, explain the variation shown. [2]
 - Polygenes;
 - Continuous variation / normal distribution / bell shaped;
 - Range seen due to the additive effects of genes / alleles;
 - Effects of environment;
- (d) How does the information in **Figure 6.2** support your answer in (c)? [1]
 - Breeding with individuals of similar genotypes;
 - Shifted the mean / variation towards either end;

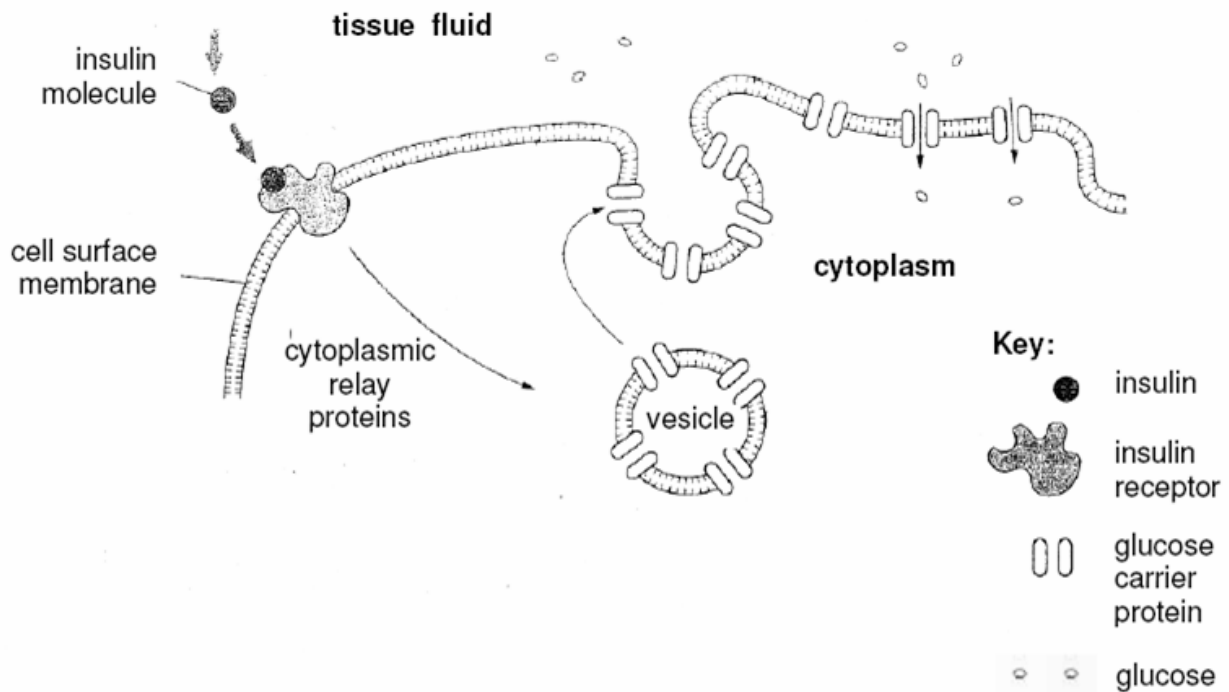
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Type 2 diabetes is a disease condition that is usually caused by the failure of target cells to respond to insulin. This is due to a defect in the insulin response mechanism of the target cells.		
Fig.		
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6.1 shows the normal sequence of cell signaling events when insulin reaches its target cell and binds



to an insulin receptor.

Fig. 6.1

(a) State the source of the insulin molecule in Fig.

6.1.

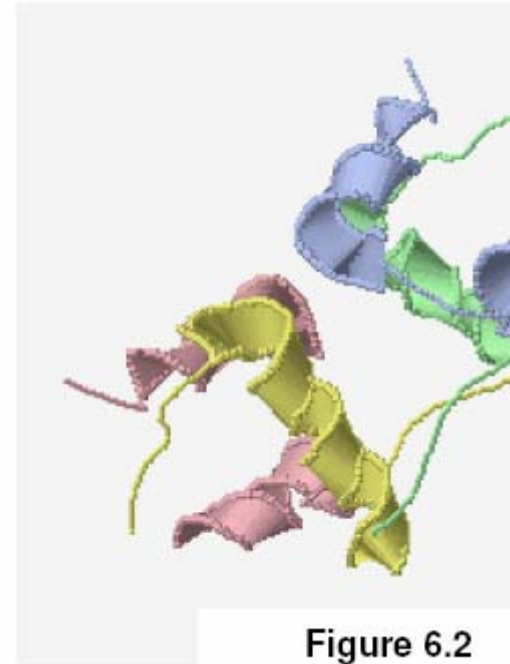
[1]

(b) The insulin receptor is a protein.

be a genetic risk factor for
Type 2 diabetes.

[1]

The structure of the
insulin receptor is
shown in Figure
6.2.



(c) With reference to Figures 6.1 and 6.2,

(i) explain how the binding of insulin to its receptor results in activation of tyrosine kinase

explain how the binding of insulin to its receptor results in an increase in the phosphorylation of serine and threonine residue side-chains of the cytoplasmic relay proteins.

(ii) describe the events

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of serine and threonine residues of cytoplasmic relay proteins

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that lead to an increase in the number of glucose channels in the cell surface membrane.

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[2]

(d) Type 2 diabetes patients are not able to regulate their blood glucose levels owing to the loss of their cells' sensitivity to insulin.

(d)

(i) If an increase in blood glucose concentration is not corrected for some time, symptoms of diabetes such as excessive thirst may occur. Suggest how excess blood glucose may cause excessive thirst.

[1]

(ii) Explain how the blood insulin level is homeostatically

regulated in a normal

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individual.

[3]

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Answer:

(a)

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Identify the source of the insulin molecule in **Figure 6.1**. [1]

Beta

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cells of the pancreatic islets of Langerhans;;
(**Reject:** b-pancreatic cells / b-islets)

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(b) The insulin receptor is a protein.

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be a genetic risk factor for Type 2 diabetes.
[1]

Harmful mutation of the **gene encoding the insulin receptor** (reject: insulin) may occur;

(Accept: reference to harmful effects of the mutation eg. alteration of specific 3-D conformation of insulin receptor or defective / non-functional / malfunctional receptor;

Reject: genetic defect / defective gene / faulty gene ie word "mutation" not mentioned.) mutant allele / gene is **transmitted to / inherited by future generations**;

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With reference to **Figures 6.1 and 6.2**,
(i)

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explain how the binding of insulin to its receptor results in activation of tyrosine kinase

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in an increase in the phosphorylation of serine and threonine residue side-chains of cytoplasmic relay proteins

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[
3
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Binding of insulin causes the **two receptor subunits / polypeptide chains / monomers / molecules** (reject: two receptors) to **form a dimer / dimerise**;
dimerisation **activates** the **tyrosine kinase** region of each subunit;

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leading to **(auto)phosphorylation of tyrosine residues** on the tail of subunits;

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Specific **relay protein(s)** in the cell **bind(s)** to specific **phosphorylated tyrosine(s)**; and becomes **activated**;

activated relay protein(s) trigger(s) **transduction pathway(s) / phosphorylation cascade**;

leading to **activation of serine and threonine kinases**; / **inhibition of serine and threonine protein phosphatases**;

in the cytoplasm. (Accept: serine and threonine kinases / protein phosphatases bind to phosphorylated tyrosine residues; and are activated; / inhibited; respectively)
(max 1.5m)

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Label the structures X and Y. [2]

X: axoplasm Y: myelin sheath

(b)

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Nervous Control
TJC Prelim 2007

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QUESTION 7

Genetic modification of crop plants can be carried out with the usage of a plant cloning vector known as the Ti plasmid. This plasmid is found in cells of the bacterium *Agrobacterium tumefaciens*, which normally lives in soil. When these bacteria infect a plant cell, a 30,000-base pair segment of the Ti plasmid, known as the T DNA sequence, separates from the plasmid and incorporates into the host cell genome. Insertion of a foreign DNA sequence into the T DNA sequence of the Ti plasmid hence allows for the insertion of that foreign DNA sequence into the plant genome. The process is shown in **Figure 4.1** below.

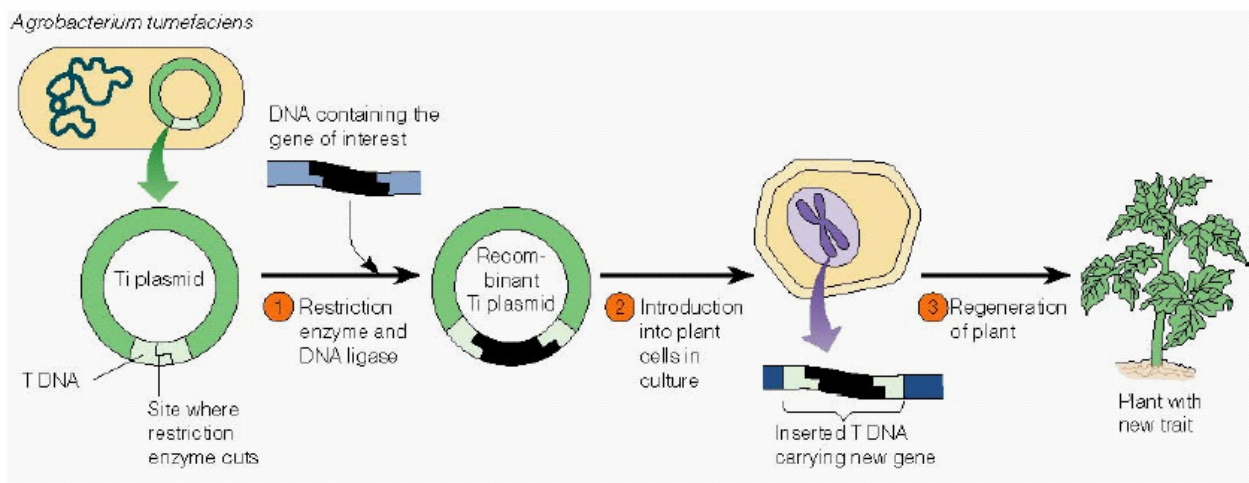


Figure 4.1

(a) Explain why it is important that the gene of interest be inserted into the T DNA sequence

of the plasmid. [2]

- (b) The usage of complementary DNA, rather than genomic DNA, is generally preferred in a situation where the host cell is a bacterial cell rather than a plant cell. Suggest why this is so. [2]

Extensive field trials have been carried out to assess the environmental effects of growing genetically modified (GM) crop plants, such as maize and oilseed rape. The crop plants have been genetically modified so that they are resistant to certain herbicides.

- (c) Why is it important to assess the environmental effects of growing GM crops? [2]

Each crop was planted in a number of different sites. One half of each site was planted with a non-GM crop and the other half was planted with the GM crop. The crops were grown for three years and during this time many different measurements were made. Measurements included the density of weeds in the crops and the numbers and types of invertebrates, such as predatory insects, living on and in the soil. **Table 4.1** shows some of the summarized results of these field trials.

Factor	GM maize compared with non-GM maize	GM oilseed rape compared with non-GM oilseed rape
Weed density	Increase	Decrease
Number of weed seeds in soil	No change	Decrease
Number of invertebrates on soil surface	Increase	Decrease

Explain the significance of Y in electrical transmission of nervous impulses.

[3]

Y acts as an electrical insulator, preventing the loss of current from the axon during electrical transmission.

In a myelinated neurone, action potentials are regenerated only at the nodes of

Ranvier / propagated by saltatory conduction,

thus increasing the speed of electrical transmission.

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Often present at the synaptic knob of many motor neurons

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Figure 2 is an electron micrograph which shows the axon of a motor neuron in transverse section.

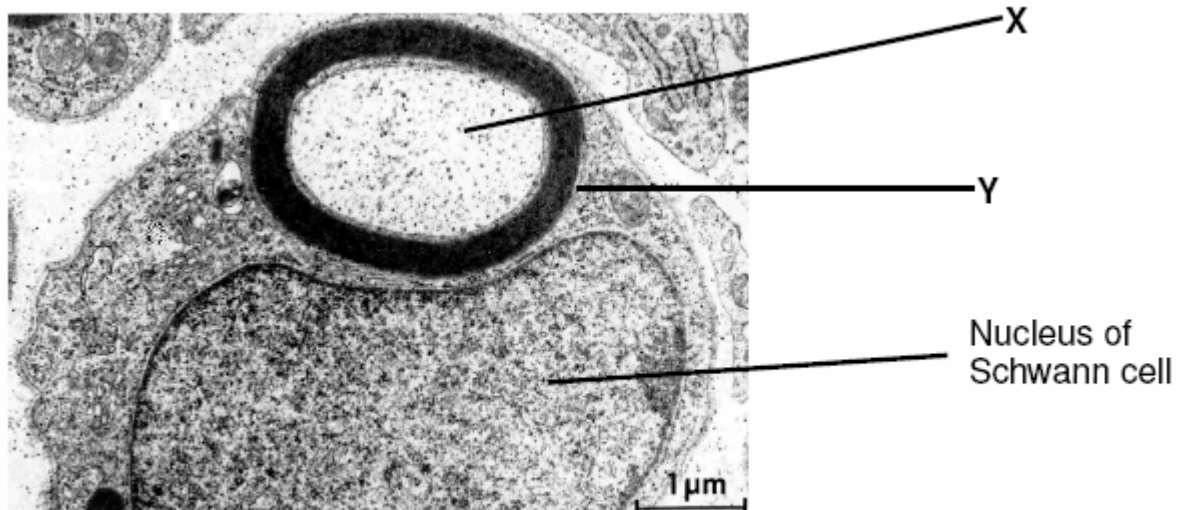


Fig. 2

(a) Label the structures X and Y.

(d) With reference to **Table 4.1**,

- (i) state which GM crop seemed to be less harmful to the environment. [1]
- (ii) give a reason for your answer in **(b)(i)**. [2]
- (iii) discuss which is a more commercially viable crop for the farmer to grow. [2]

The scientists further investigated the transfer of the introduced gene from the GM maize and oilseed rape plants to other plants.

(e) Suggest how the transfer could have taken place. [1]

[Total: 12]

- (a) Explain why it is important that the gene of interest be inserted into the T DNA sequence of the plasmid. [2]

T DNA is inserted into the plant genome; Hence allowing for **foreign gene** within it to **also be inserted into the plant genome;** **Stable** expression (vs transient expression); **Passed on to all daughter cells** of the transformed cell;

(b) The usage of complementary DNA, rather than genomic DNA, is generally preferred in a situation where the host cell is a bacterial cell rather than a plant cell. Suggest why this is so. [2]

complementary DNA does not contain introns; use of genomic DNA necessitates the **removal of introns**; **which could not be carried out by a bacterial cell**; as bacterial cell **lacks splicing machinery**; not having any introns within its own genes; **Functional** foreign protein can thus be expressed by bacterial host; since **cDNA contains only coding sequence**;

(c) Why is it important to assess the environmental effects of growing GM crops? [2]
Discussion of ecological implications e.g.

possibility of **transfer of introduced herbicide resistance gene from GM crops to wild relatives**; through **cross-pollination**; resulting in development of “**superweeds**”; which are difficult to eradicate; need to assess possible effects of transgene products, like toxins, on other organisms; e.g. endangered wild species;

(d) With reference to **Table 4.1**,

(i) state which genetically modified crop seemed to be less harmful to the environment. [1]
GM maize;;

(ii) give an explanation for your answer in (d)(i). [2]
Weeds / surface invertebrates increase for GM maize compared to GM oilseed rape;; indicating that the presence of the proteins to confer resistance to herbicides have not hindered the growth and proliferation of other non-threatening organisms like surface invertebrates;;

(iii) discuss which is a more commercially viable crop for the farmer to grow. [2]

GM oilseed rape;;
due to **decreased weed density** or **decreased number of invertebrates pests** (relate these to **increased crop yield**);;

OR

Lower crop losses due to reduced competition for nutrients with weeds; And due to less damage by **reduced numbers of invertebrate pests**;

OR

Lower costs of herbicides to kill off weeds; and **costs of pesticides to kill invertebrate pests**;

(c) Suggest how the transfer could have taken place. [1]

Transgene present in pollen grains; Through **pollen transfer**; from **GM maize / GM oilseed rape** to other plants resulting in **pollination**; and **fertilization**;

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ions are taken into the neurone via active transport.

ATP is required for the action of Ca²⁺ pumps which actively transport Ca²⁺ ions out

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1]

X:

Y:

(b) Explain the significance of Y in electrical transmission of nervous impulses. [3]

Often present at the synaptic knob of many motor neurons is a large number of mitochondria.

(c)(i) State two uses for the ATP produced by these mitochondria in the synaptic knob of a motor neurone.

[2]

(c)(ii) Suggest why during a period of intense nervous activity, the metabolic rate of a nerve cell increases.

[2]

(d) Predict the effect on an action potential when the external concentration of sodium ions are lowered.

[2]

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Answers:

Figure 2 is an electron micrograph which shows the axon of a motor neurone in transverse section.

(a)

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are a large number of mitochondria.

(c)(i)

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State two uses for the ATP produced by these mitochondria in the synaptic knob of a motor

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neurone. [2]

Any two:

ATP is required for the Na⁺/K⁺ pump to work. Na⁺ ions are pumped out while K⁺

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of the neurone.

ATP is used for the transport / movement of synaptic vesicles towards the

membrane of the synaptic knob for the release of neurotransmitter.

ATP used for the absorption / endocytosis of choline and acetate back into the

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(c)(ii) Suggest why during a period of intense nervous activity, the metabolic rate of a nerve cell increases.

[2]

During intense activity, more action potentials are generated;
Increased activities of Na⁺ - K⁺ pumps needing more ATP

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- (d) Suggest the effect on an action potential when the external concentration of sodium ions are lowered.
[2]

Less sodium ions enter the neurone;
Hence lower action potential

- (d) (i)
High Na^+ concentration outside the cell and a high K^+ concentration inside the cell;
Quote any cell's values –

Location	Ions	Ions concentration in			
		Frog muscle	Human r.b.c	Squid axon	<i>Chaetomorpha</i> (marine alga)
Outside the cell	Na^+	114	150	400	500
	K^+	14	14	14	14
Inside the cell	Na^+	14	14	42	42
	K^+	135	150	300	400

[Each point → 1 mark]

- (ii)
Cell membrane;
 Na^+ and K^+ ion channels/ Na^+/K^+ pump;

[Each point → ½ marks]

- (iii)
Maintains the resting membrane potential to generate nerve impulse;
OR
Contributes to the electrochemical gradient to establish the resting membrane potential;

[Each point → 1 mark]

- (c)
Action potential arrives at the synaptic knob;
Depolarization increases membrane permeability to Ca^{2+} ;
opening of voltage gated calcium channels and Ca^{2+} influx;
 Ca^{2+} influx causes synaptic vesicles to fuse with presynaptic membrane and discharge neurotransmitters by exocytosis;

Underlined words should appear in student's answer.

[Each point → ½ marks]

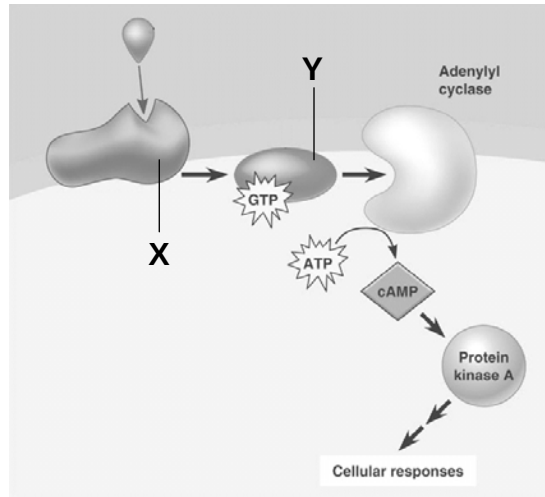
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Cell Signaling
SAJC Prelim 2007
QUESTION 8

Certain microbes cause disease by disrupting G-protein signaling pathways. The cholera bacterium, *Vibrio cholerae*, present in water contaminated with human feces colonizes the small intestine and produces a toxin that modifies a G protein.

The toxin results in a high concentration of cAMP which causes the intestinal cells to secrete large amounts of water and salts into the intestines, leading to profuse diarrhea and death from loss of water and salts.



(a) Name molecules X and Y.

X Y[1]

(b) What is the general name given to small molecules or ions that function like cAMP in signaling pathways?

.....[1]

(c) What is the significance of a phosphorylation cascade in cell signaling?

.....

[2]

(d) State the function of adenyl cyclase.

.....[1]

(e) Suggest how the toxin leads to an increase in cAMP concentration.

.....

[3]

[Total: 8]

(a) Name molecules X and Y. [1]

[½ mark per pt]

X = G-protein linked receptor

Y = G-protein

(b) What is the general name given to small molecules or ions that function like cAMP in signaling pathways? [1]
 second messengers

(c) What is the significance of a phosphorylation cascade in cell signaling? [2]

signal amplification

specificity of cell signaling

ref. cross-talk (interaction between different signaling pathways)

(d) State the function of adenylyl cyclase. [1]

convert ATP to cAMP

(e) Suggest how the toxin leads to an increase in cAMP concentration. [3]

modified G protein unable to hydrolyze GTP to GDP

remains stuck in its active form

adenylyl cyclase is continuously stimulated to make cAMP

Page Break

NADP/ NAD as

- Electron/hydrogen carrier or coenzymes;
 - Carry hydrogen atoms from one compound to another;
- [max1, ½ m each]

NADP in photosynthesis

- Final electron acceptor in non-cyclic photophosphorylation;
 - Drives photolysis of water;
 - $2\text{H}^+ + 2\text{e}^- + \text{NADP} \rightarrow \text{NADPH}$ or reduced NADP/ hydrogen ions from splitting of water combine with NADP to form NADPH;
 - NADPH passes to Calvin cycle/ light independent reactions;
 - in the stroma;
 - NADPH reduces glycerate biphosphate to triose phosphate and gets reoxidized back to NADP;
- [max2, ½ m each]

NAD in respiration

Aerobic respiration:

Glycolysis: Accept H^+ from triose phosphate, which is oxidized to pyruvate;

Forms 2 NADH for each glucose molecule;

Link reaction: Accept H^+ from pyruvate, which is converted to acetyl CoA;

Forms 2 NADH for each glucose molecule;

Krebs cycle: Accept H^+ from intermediates in the regeneration of oxaloacetate from citrate;

Forms 6 NADH for each glucose molecule;

Link rxn and Krebs cycle occurs in the mitochondrial matrix

Carry hydrogen/ electrons to ETC;

Oxidative phosphorylation/ transfer of electrons down the ETC to release ATP;

1 NADH yields 3 ATP;

NAD reoxidised;

Anaerobic respiration:

NAD regenerated when electrons from NADH transferred to pyruvate/ ethanal;

NAD reused in glycolysis to generate 2 ATP;

[max5, ½ m each]

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NADP/ NAD as

- Electron/hydrogen carrier or coenzymes;
 - Carry hydrogen atoms from one compound to another;
- [max1, ½ m each]

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 - Drives photolysis of water;
 - $2\text{H}^+ + 2\text{e}^- + \text{NADP} \rightarrow \text{NADPH}$ or reduced NADP/ hydrogen ions from splitting of water combine with NADP to form NADPH;
 - NADPH passes to Calvin cycle/ light independent reactions;
 - in the stroma;
 - NADPH reduces glycerate biphosphate to triose phosphate and gets reoxidized back to NADP;
- [max2, ½ m each]

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Feature	photophosphorylation	oxidative phosphorylation
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Carry hydrogen/ electrons to ETC;
 Oxidative phosphorylation/ transfer of electrons down the ETC to release ATP;
 1 NADH yields 3 ATP;
 NAD reoxidised;

Anaerobic respiration:

NAD regenerated when electrons from NADH transferred to pyruvate/ ethanal;
 NAD reused in glycolysis to generate 2 ATP;
 [max4, ½ m each]

Coenzyme NAD and NADP are reduced, carry electrons and protons to the electron transfer chain and are regenerated;
 NAD is formed in glycolysis, link reaction and Krebs's cycle by oxidation;
 in the mitochondrial matrix;
 And then used in oxidative phosphorylation;
 NADP location is in the stroma of the chloroplast;
 NADP is reduced in the light reaction as the final electron acceptor;
 And then was used in the Calvin cycle as a reducing agent;

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Organelle	Chloroplast	Mitochondria;
Location	Thylakoid membrane	Inner Mitochondrial membrane;
Proton reservoirs	Thylakoid space	Intermembrane space;
Energy conversion	Light to chemical	Chemical to chemical;
Role of coenzymes	NADP is reduced in the light rxn as final electron acceptor	NAD is reduced in glycolysis, link rxn and Kreb's cycle;
Sources of electrons	Water	NADH and FADH;
Roles of O ₂	By-product of photolysis	Final electron acceptor;
Roles of water	Photolysis: To produce H ⁺ which will then accumulate in the thylakoid space and electrons to replace the electron lost at the primary reaction centre	By-product of O ₂ as the final electron acceptor;

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Feature	photophosphorylation	oxidative phosphorylation
Organelle	Chloroplast	Mitochondria;
Location	Thylakoid membrane	Inner Mitochondrial membrane;
Proton reservoirs	Thylakoid space	Intermembrane space;
Energy conversion	Light to chemical	Chemical to chemical;
Role of coenzymes	NADP is reduced in the light rxn as final electron acceptor	NAD is reduced in glycolysis, link rxn and Kreb's cycle;
Sources of electrons	Water	NADH and FADH;
Roles of O ₂	By-product of photolysis	Final electron acceptor;
Roles of water	Photolysis: To produce H ⁺ which will then accumulate in the thylakoid space and electrons to replace the electron lost at the primary reaction centre	By-product of O ₂ as the final electron acceptor;

[Any 7 – 1 mark each]

Statements must be comparative statements.

Explain the effect of carbon dioxide and light as limiting factors on the rate of photosynthesis.

CO₂ is a raw material for the calvin cycle

Its increased concentration increased the rate of photosynthesis

Its level is low (0.04%) in the atmosphere and therefore plays a major role in limiting photosynthesis

(c) Light intensity is also important in the light dependent stage to excite electrons

Chl a absorbs max light at 680nm while chl b absorbs max light at 480nm

Compensation point is the light intensity when rate of photosynthesis and respiration is the same and it is reached at low light intensity

Is therefore rarely limiting during daylight hours

High light intensity also opens the stomata

[6]

Photosynthesis

Photosystems / antennae complex / chlorophyll in thylakoid membranes;
 ETC, stalked particles (ATP synthase) in thylakoid membranes;
 Thylakoid membranes are ion-impermeable / ref. to H^+ gradient;
 Ref. to thylakoid membrane separating compartments (compartmentalization);
 Ref. to double membrane / envelope;
 Ref. to enzymes and Calvin cycle in the stroma;

[max 3]

Respiration

Folding of cristae to increase surface area so that there are more stalked particles;
 More ETC for oxidative phosphorylation;
 ETC and stalked particles in inner membrane / cristae;
 Inner membrane ion impermeable / ref. to proton gradient;
 Ref. to intermembrane space acting as reservoir of H^+ ;
 Ref. to outer membrane as a barrier;
 Ref. to enzymes and Krebs cycle in matrix

[max 3]

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